📘 HOTBEE Commerce Platform V1.0

Complete Software Architecture & Developer Implementation Guide

Objective

Build a fully automated Print-on-Demand eCommerce platform using **Next.js + Supabase + Razorpay + Qikink** without Shopify and without AI.

Technology Stack

Frontend

- Next.js (App Router)
- React
- TypeScript
- Tailwind CSS
- shadcn/ui
- Lucide Icons

Backend

- Supabase Edge Functions (No FastAPI)

Database

- Supabase PostgreSQL

Authentication

- Supabase Auth

Storage

- Supabase Storage

Payment

- Razorpay

Print On Demand

- Qikink API

Hosting

- Netlify

Analytics

- Google Analytics
- Meta Pixel

Email

- Resend (Free Tier)

PART 1 — Business Architecture

Business Flow

Customer

↓

Website

↓

Product

↓

Cart

↓

Checkout

↓

Payment

↓

Qikink

↓

Customer

Admin Flow

Brother Login

↓

Dashboard

↓

Orders

↓

Products

↓

Revenue

↓

Shipping

↓

Customers

↓

Coupons

↓

Reviews

Customer Flow

Landing Page

↓

Browse Products

↓

Product Details

↓

Cart

↓

Checkout

↓

Payment

↓

Order Confirmation

↓

Track Order

↓

Review Product

PART 2 — System Architecture

```
Customer

↓

Next.js Website (Netlify)

↓

Supabase

├── PostgreSQL

├── Auth

├── Storage

└── Edge Functions

↓

Razorpay

↓
```

```
Qikink

↓

Courier

↓

Customer
```

PART 3 — Developer Roadmap

Phase 1 — Project Initialization

- Create GitHub Repository
- Next.js
- React
- TypeScript
- Tailwind CSS
- shadcn/ui
- ESLint
- Prettier
- Folder Structure
- Environment Variables
- Deploy Starter to Netlify

Phase 2 — Supabase Setup

Step-by-Step

- Create Supabase Account
- Create Project
- Select Region
- Get Project URL
- Get Anon Key
- Get Service Role Key
- Configure Authentication
- Configure Storage
- Configure Edge Functions
- Configure RLS
- Configure Policies

Phase 3 — Database Design

Tables

- Users
- Admin
- Products
- Product Images
- Categories
- Collections
- Customers
- Addresses
- Orders
- Order Items
- Payments
- Coupons
- Reviews
- Wishlist
- Cart
- Shipping
- Notifications
- Contact Messages

Includes:

- ER Diagram
- SQL Scripts
- Foreign Keys
- Indexes
- Relationships

Phase 4 — Qikink Integration

Developer Guide

- Create Qikink Account
- Generate API Credentials
- Understand Product IDs
- Product Creation
- Product Sync
- Create Order API
- Order Status API
- Tracking API
- Cancel Order API
- Error Handling
- Wallet Handling

Phase 5 — Razorpay Integration

Developer Guide

- Create Merchant Account
- Complete KYC
- Enable Test Mode
- Enable Live Mode
- Generate API Keys
- Create Orders
- Verify Payments
- Verify Signature
- Configure Webhooks
- Handle Failed Payments
- Handle Refunds

Phase 6 — Supabase Edge Functions

Implement

- Create Razorpay Order
- Verify Razorpay Payment
- Save Order
- Create Qikink Order
- Update Order Status
- Send Confirmation Email
- Update Analytics

Phase 7 — Frontend Development

Public Pages

- Landing Page
- Collections
- Product Listing
- Product Details
- Search
- Category
- Wishlist
- Cart
- Checkout
- Payment Success

- Payment Failed
- Contact
- About
- Privacy Policy
- Terms & Conditions
- Refund Policy
- Shipping Policy
- 404 Page

Customer Pages

- Login
- Register
- Forgot Password
- My Profile
- Address Book
- Order History
- Order Details
- Track Order
- Wishlist
- Notifications

Phase 8 — Admin Dashboard

Authentication

- Secure Admin Login

Dashboard

- Today's Orders
- Monthly Revenue
- Pending Orders
- Delivered Orders
- Cancelled Orders

Products

- Add Product
- Edit Product
- Delete Product

- Categories
- Collections

Orders

- View Orders
- Payment Status
- Qikink Status
- Tracking Number
- Invoice Download

Customers

- Customer Details
- Address
- Order History

Coupons

- Create Coupon
- Edit Coupon
- Disable Coupon

Reviews

- Approve Reviews
- Delete Reviews

Settings

- Brand Information
- Shipping Charges
- Tax Settings
- Contact Details

Phase 9 — Deployment

- Netlify
- Supabase
- Environment Variables
- SSL
- Production Build
- Performance Optimization

Phase 10 — Testing

Payment Testing

- Successful Payment
- Failed Payment
- Duplicate Payment

Qikink Testing

- Create Order
- Track Order
- Cancel Order

User Testing

- Register
- Login
- Checkout
- Wishlist
- Reviews
- Coupons

Production Checklist

- Performance
- Security
- Mobile Responsive
- Cross Browser
- Accessibility

PART 4 — Brother Setup Guide

Step 1

Create Gmail

Step 2

Create Qikink Account

Provide:

- Login Credentials
- Store Name
- API Key
- API Secret

Step 3

Create Razorpay

Complete:

- PAN
- Aadhaar
- Bank Verification
- KYC

Provide:

- Key ID
- Secret
- Webhook Secret

Step 4

Create Supabase

Provide:

- Project URL

- Anon Key
- Service Role Key

Step 5

Brand Assets

- Logo
- Favicon
- Banner
- Product Images

Step 6

Product Setup

- Upload Products
- Selling Price
- Categories
- Collections

Step 7

Wallet Setup

- Initial Wallet Recharge
- Recharge Process
- Wallet Monitoring

Step 8

Testing

- Test Payment
- Test Order
- Test Tracking

Step 9

Go Live

- Instagram Bio
- Marketing
- Customer Testing

PART 5 — Folder Structure

```
hotbee/

app/
components/
features/
hooks/
lib/
services/
supabase/
edge-functions/
store/
types/
utils/
constants/
public/
styles/
middleware.ts
```

PART 6 — Environment Variables

```
NEXT_PUBLIC_SUPABASE_URL=

NEXT_PUBLIC_SUPABASE_ANON_KEY=

SUPABASE_SERVICE_ROLE_KEY=

QIKINK_API_KEY=

QIKINK_API_SECRET=

RAZORPAY_KEY_ID=

RAZORPAY_SECRET=

RAZORPAY_WEBHOOK_SECRET=

RESEND_API_KEY=

NEXT_PUBLIC_GA_MEASUREMENT_ID=

NEXT_PUBLIC_META_PIXEL_ID=
```

PART 7 — Features

Customer

- Product Listing
- Categories
- Collections
- Search
- Wishlist
- Cart
- Checkout
- Razorpay Payment
- Order History
- Track Order
- Reviews
- Coupons
- Profile
- Address Management
- Email Confirmation

Admin (Brother)

- Dashboard
- Products
- Categories
- Collections
- Orders
- Customers
- Coupons
- Reviews
- Revenue Reports
- Shipping Status
- Qikink Status
- Wallet Reminder
- Analytics
- Settings

PART 8 — APIs

Supabase

- Auth
- Database
- Storage
- Edge Functions

Razorpay

- Orders
- Payments
- Webhooks

Qikink

- Products
- Create Order
- Track Order
- Cancel Order

PART 9 — Security

- Row Level Security
- JWT Authentication
- Secure Environment Variables
- Razorpay Signature Verification
- Qikink Secret Protection
- Webhook Verification
- Input Validation
- API Rate Limiting
- HTTPS
- Secure Headers

PART 10 — Future Roadmap

Version 2

- Custom Domain
- Referral System
- Loyalty Points
- Gift Cards
- Inventory Alerts
- Multi-language Support
- Multi-currency Support
- Multiple POD Providers

📖 Final Deliverable

This becomes your **HOTBEE Commerce Platform Developer Bible** (~150–180 pages) containing:

- ☑ Business Architecture
- ☑ Complete UI/UX Planning
- ☑ Database Design
- ☑ SQL Scripts
- ☑ Supabase Setup
- ☑ Razorpay Setup
- ☑ Qikink Setup
- ☑ Edge Functions
- ☑ Next.js Implementation
- ☑ Admin Dashboard
- ☑ Customer Website
- ☑ Deployment Guide
- ☑ Security Guide
- ☑ Testing Guide
- ☑ Brother Operations Manual